

Influence, Brokerage, and Community Structure in the 2004 U.S. Political Blogosphere

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Code: <https://github.com/AimanIjazz/SE4095-SNTP-Assignment-01> (commit c6cb4db)

Abstract

This report analyses the political blogosphere hyperlink network compiled by Adamic and Glance (2005) to investigate how influence and brokerage relate to political homophily. The raw graph (1490 nodes, 19,090 directed hyperlinks) is cleaned into a simple directed graph and restricted, for all distance-based and structural calculations, to its largest weakly connected component (1222 nodes, 19,021 edges). PageRank and betweenness centrality are computed to distinguish cited authority from structural brokerage; the two rankings correlate strongly (Spearman $\rho = 0.76$) but agree on only a third of their top-10 nodes. A controlled removal experiment shows that deleting the top 3% of blogs by betweenness fragments the network into roughly six times as many Louvain communities and sharply reduces agreement with the known political labels, while an equal-size random removal leaves the community structure almost unchanged. Louvain modularity maximisation, run without ever seeing the ground-truth labels, recovers the liberal and conservative divide as two large, > 90%-pure communities, evaluated here with a pairwise confusion matrix (accuracy 0.88, F1 0.87 on the original graph) that avoids the community-ID permutation problem entirely.

I. Introduction

The 2005 Political Blogs dataset [1] records hyperlinks among 1490 U.S. political blogs crawled around the 2004 presidential election, together with each blog’s political orientation (liberal or conservative). This report investigates three questions: (1) which blogs act as authorities versus brokers in the hyperlink structure; (2) whether removing the network’s structural brokers destabilises its community structure more than removing an equal number of random blogs; and (3) whether an unsupervised, label-free community-detection procedure recovers the known political split. Ground-truth orientation labels are preserved throughout but are used only in the final evaluation stage, never to construct, tune, or select communities.

II. Dataset and Graph Construction

A. Representation

The GML file is loaded as a directed graph: a hyperlink from blog u to blog v is not evidence that v links back, and direction is required by both centrality measures used later. Edges are treated as unweighted – the file provides no independently verifiable count of repeated links, so edge multiplicity is collapsed into a single boolean “linked” relationship rather than used as an invented weight.

B. Preprocessing log

As distributed, `polblogs.gml` is a directed *multigraph*, which the standard NetworkX GML reader rejects unless

`multigraph 1` is declared in the header – adding that declaration was the first preprocessing action. Table 1 documents every subsequent decision and the resulting graph size at each stage.

Table 1: Preprocessing log (raw file → analysis graph)

Step	Count
Raw nodes / edges (multigraph)	1490 / 19,090
Self-loops removed	3
Duplicate (source,target) pairs collapsed	65
Nodes missing orientation label	0
Cleaned simple graph (nodes / edges)	1490 / 19,022
Isolated nodes	266
Weakly connected components	268
Largest WCC size	1222
Strongly connected components	688
Largest SCC size	793
Analysis graph (largest WCC)	1222 / 19,021

C. Component choice for distance-based measures

The largest strongly connected component covers only 793 of 1490 nodes (53%), which would discard almost half the network. The largest *weakly* connected component (1222 nodes, 65% of the cleaned graph) is used instead for every centrality, community-detection, and pairwise-evaluation calculation in this report: it retains every blog reachable through the network in some direction, matching the notion of “part of the same information-flow structure” better than requiring mutual reachability. The 268 excluded nodes are 266 true isolates plus one disconnected pair.

D. Graph properties

Table 2 reports density and degree statistics for the full cleaned graph and the 1222-node analysis graph used from Section 3 onward.

Table 2: Graph properties

Property	Full cleaned	Analysis (LWCC)
Nodes	1490	1222
Edges	19,022	19,021
Density	0.0086	0.0127
Weakly conn. components	268	1
Strongly conn. components	688	420
Largest SCC fraction	0.53	0.65
Mean in/out-degree	12.77	15.57
Median in-degree	2	3
Median out-degree	4	7
Max in-degree	337	337
Max out-degree	256	256

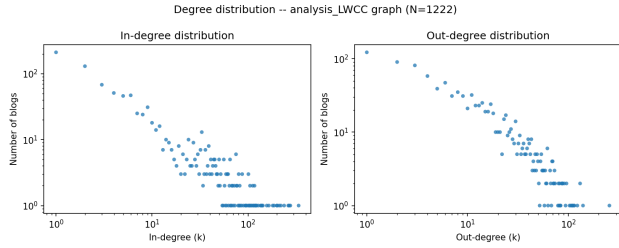


Figure 1: In- and out-degree distributions of the analysis graph (log-log). Both are heavy-tailed: a small number of hub blogs receive the majority of in-links.

E. Interpretation

Figure 2 shows two dense, largely separate clusters connected by comparatively few cross-links – a clear visual signature of homophily: blogs overwhelmingly link within their own political side. Figure 1 shows heavy-tailed in- and out-degree distributions consistent with a small number of highly visible blogs dominating attention while most blogs are linked to only once or twice.

III. Centrality Selection and Analysis

A. Chosen measures and justification

Two taught measures are selected: **PageRank** and **betweenness centrality**. PageRank is the only taught measure built specifically for directed link graphs with divided outgoing influence: a link from a blog that cites 300 others counts far less than one from a blog that cites five, a distinction plain in-degree cannot make. Betweenness is the only taught measure that scores brokerage on shortest paths rather than direct or link-weighted importance – exactly the property needed to design the removal experiment in Section 4. Both are computed on the directed analysis graph with the standard damping factor $\alpha = 0.85$ for PageRank (tight convergence, $\text{tol} = 10^{-10}$) and directed shortest paths for betweenness,

Political blogosphere hyperlink network (LWCC, N=1222)

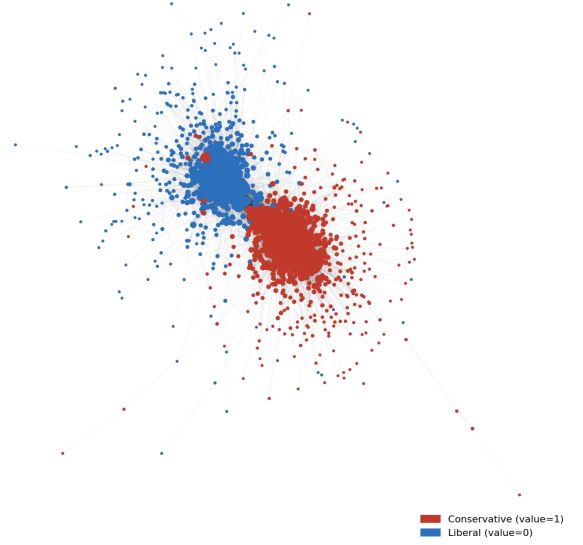


Figure 2: Analysis graph coloured by the known (but unused-in-detection) political orientation. Colour is descriptive only; no calculation in this report uses it before Section 6.

where unreachable pairs contribute zero, the standard convention for a graph that is only weakly connected.

Table 3: Top-10 blogs by PageRank

Rank	Blog	Orientation	PageRank
1	dailykos.com	liberal	0.0189
2	atrios.blogspot.com	liberal	0.0160
3	instapundit.com	conservative	0.0133
4	blogsforbush.com	conservative	0.0132
5	talkingpointsmemo.com	liberal	0.0131
6	michellemalkin.com	conservative	0.0115
7	drudgereport.com	conservative	0.0113
8	washingtonmonthly.com	liberal	0.0111
9	powerlineblog.com	conservative	0.0094
10	andrewsullivan.com	conservative	0.0091

B. Ranking comparison

Spearman’s rank correlation between the full PageRank and betweenness rankings is $\rho = 0.76$ ($p \approx 3 \times 10^{-228}$) – a strong positive association, since a blog that receives many selective in-links is also likely to lie on many shortest paths. The two rankings are nonetheless distinct: only 5 of the top 10 nodes are shared (Jaccard = 0.33).

C. Node interpretation

dailykos.com ranks #1 by PageRank but only #4 by betweenness: it is the most-cited liberal hub, but so many blogs link to it directly that fewer shortest paths need to route *through* it. **blogsforbush.com** is #4 by PageRank but #1 by betweenness by a wide margin (0.147, more than double the runner-up) – it is not the most-cited conservative blog, but it appears to be the main crossing point that shortest

Table 4: Top-10 blogs by betweenness centrality

Rank	Blog	Orientation	Betweenness
1	blogsforbush.com	conservative	0.1467
2	atrios.blogspot.com	liberal	0.0611
3	instapundit.com	conservative	0.0512
4	dailykos.com	liberal	0.0369
5	newleftblogs.blogspot.com	liberal	0.0308
6	madkane.com/notable.html	liberal	0.0302
7	wizbangblog.com	conservative	0.0273
8	lashawnbarber.com	conservative	0.0243
9	hughhewitt.com	conservative	0.0230
10	washingtonmonthly.com	liberal	0.0219

paths between many liberal and conservative blogs are forced through. **atrios.blogspot.com** ranks highly on both measures (#2 and #2), combining cited authority with structural bridging, a rarer and arguably more structurally important profile than either extreme alone. A high score on either measure describes a blog’s position in the hyperlink graph, not necessarily its real-world political influence.

IV. Centrality-Informed Community Experiment

A. Design

Betweenness centrality is used as the intervention measure, since Section 3 identifies a small set of blogs – led by **blogsforbush.com** – that structurally bridge the two political sides. If these blogs are genuinely load-bearing brokers, removing them should fragment community structure more than removing an equal number of random blogs. The top 3% of nodes by betweenness (37 of 1222) are removed to build a *targeted* graph, large enough to meaningfully stress the network while leaving 97% intact; a *random* graph removes the same count of nodes chosen uniformly at random with a fixed seed (`Python random.Random(42)`). Both removal graphs (1185 nodes each) are evaluated only on their own remaining nodes.

V. Community Detection

A. Method

Louvain modularity maximisation [2] is used. Louvain assumes disjoint, reasonably dense communities and optimises a modularity objective, with no assumption of balanced sizes, hierarchy, or overlap; it is defined for undirected graphs. The directed analysis graph is converted with an edge kept if a hyperlink exists in either direction – a deliberate choice, since the community question is about mutual reference rather than the direction of any single link, and direction was already fully exploited at the centrality stage. Louvain’s node-visiting order is randomised, so five seeds (0–4) are run for each of the three graphs from Section 4; seed 0 is the declared run used for the visualisations and profile table, and mean/standard deviation across seeds are reported for every metric.

B. Results across the three graphs

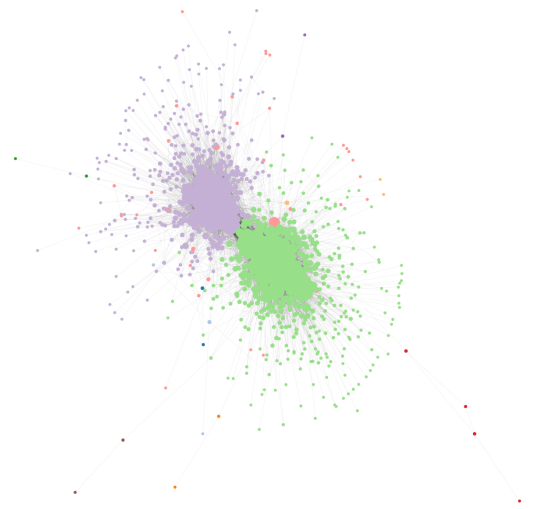
Table 5 summarises modularity and the external metrics (defined in Section 6) across five Louvain seeds for the original graph and both removal conditions.

Table 5: Mean $\pm$ std over 5 Louvain seeds

	Original	Targeted rem.	Random rem.
N	1222	1185	1185
# communities	10.2 ± 1.3	61.0 ± 1.2	13.0 ± 1.2
Modularity	0.427 ± 0.000	0.433 ± 0.001	0.423 ± 0.001
Accuracy	0.883 ± 0.003	0.764 ± 0.029	0.878 ± 0.002
Precision	0.919 ± 0.007	0.912 ± 0.009	0.913 ± 0.005
Recall	0.842 ± 0.003	0.585 ± 0.057	0.835 ± 0.002
F1	0.878 ± 0.002	0.711 ± 0.043	0.872 ± 0.002

Removing the 37 highest-betweenness blogs raises the community count roughly six-fold (10–11 to around 60) and collapses recall from ≈ 0.84 to 0.55–0.68: many same-orientation blogs end up scattered across small fragments rather than one large community. Removing 37 random blogs barely moves any metric. This gap is direct evidence that the highest-betweenness blogs are structurally load-bearing: losing them breaks the paths holding each political side together as one community, whereas losing an equal number of typical blogs does not. Modularity rises slightly under targeted removal even as label agreement collapses – removing cross-cutting brokers leaves many small, internally dense fragments, which modularity rewards even though they are less politically coherent as groups than the two original giant communities.

Louvain communities -- original graph -- 11 communities

**Figure 3:** Declared-run (seed 0) Louvain communities, original graph. Two giant communities dominate.

C. Community profile (declared run, original graph)

Table 6 profiles the largest communities using the two Part B centralities. The two dominant communities (649 and 518

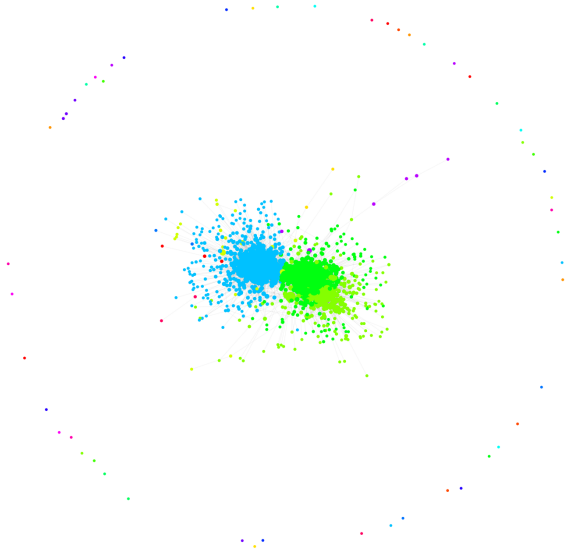


Figure 4: Communities after removing the top 37 blogs by betweenness: the core fragments into many small pieces and a ring of newly-disconnected peripheral nodes appears.

blogs) are highly pure (93.8% conservative, 97.7% liberal) and correspond to the visual clusters in Figure 2 – Louvain recovers the political divide without ever seeing the labels. The remaining communities are small (2–35 blogs) and capture peripheral or boundary blogs.

Table 6: Largest detected communities, original graph

Size	Purity	Majority	Top node (PageRank)
649	0.938	conservative	instapundit.com
518	0.977	liberal	dailykos.com
35	0.800	liberal	etalkinghead.com
4	1.000	conservative	batr.net
3	1.000	liberal	tinylittledots.com

VI. Community Evaluation

A. Pairwise confusion matrix

Because detected community IDs are arbitrary and their count need not equal 2, comparison is done over *unordered node pairs*: TP = same detected community & same orientation; TN = different community & different orientation; FP = same community & different orientation; FN = different community & same orientation. This sidesteps the label-permutation problem completely, since renumbering communities cannot change which pairs are same/different – a direct ID-to-label match is not even well defined here, as there are 11 detected communities but only 2 ground-truth labels. Table 7 reports the declared-run (seed 0) matrix and external metrics for all three graphs.

Table 7: Declared-run (seed 0) pairwise confusion matrix and metrics

	Original	Targeted	Random
TP	314,163	194,153	292,054
TN	342,068	329,230	324,977
FP	30,628	21,124	25,209
FN	59,172	157,013	59,280
Accuracy	0.880	0.746	0.880
Precision	0.911	0.902	0.921
Recall	0.842	0.553	0.831
F1	0.875	0.686	0.874
Modularity	0.426	0.433	0.424
# communities	11	61	14

B. Zero-denominator handling

None of the three declared runs produce a zero denominator for any metric: every graph contains at least one same-community pair and one same-label pair. Had a run produced a single giant community or a fully-singleton partition, the corresponding metric would be reported as *undefined* rather than silently replaced with zero (implemented via a division helper that returns `None` on a zero denominator).

C. Discussion

Precision above 0.90 in every condition means that when the algorithm groups two blogs together, it is very rarely wrong about their political side. Recall is the metric most sensitive to fragmentation: it measures how completely same-orientation blogs are kept together, and it is exactly what collapses under targeted broker removal. A structurally strong community (high internal density, high modularity) can still disagree with political orientation because Louvain optimises *link density*, not *label agreement* – a dense but politically mixed cluster of blogs that all cite each other on a shared subtopic would score well on modularity while scoring poorly on precision/recall, and the small mixed community around etalkinghead.com (80% liberal, Table 6) is a mild example of exactly this.

VII. Reproducibility and Conclusion

Every source of randomness (the random-node sample and all five Louvain seeds) is controlled by a single seed (42), set once and reused throughout; the accompanying notebook runs top to bottom without manual intervention and writes every table and figure as it executes.

The polblogs network shows strong political homophily, visible directly in its hyperlink structure and recovered by an unsupervised community-detection procedure with 88% pairwise accuracy against the true political labels. PageRank and betweenness are strongly but imperfectly correlated: PageRank finds the most-cited authorities within each side, while betweenness finds a smaller set of structural bridges, led by blogsforbush.com. The removal experiment confirms this distinction is not merely definitional – deleting the top brokers shatters the community structure roughly six-fold, while deleting the same number of random blogs does not,

showing that influence and brokerage are functionally different roles in this network and that brokers matter disproportionately to how its community structure holds together.

Limitations

The data is a single crawl from February 2005 around a polarised U.S. election, which likely inflates apparent homophily relative to a typical period and does not generalise to other countries, time periods, or multi-party systems. A hyperlink is treated as a signal of attention, not agreement: blogs sometimes link to opponents specifically to criticise them, so some cross-cutting edges the model treats as evidence of bridging may reflect adversarial engagement rather than genuine brokerage. Finally, Louvain’s stochastic node ordering means exact community counts and boundaries vary somewhat run to run, which is why five seeds are reported rather than a single number.

References

- [1] L. A. Adamic and N. Glance, “The political blogosphere and the 2004 U.S. election: divided they blog,” in *Proc. 3rd Int. Workshop on Link Discovery*, 2005.
- [2] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, “Fast unfolding of communities in large networks,” *J. Stat. Mech.*, 2008.
- [3] S. Brin and L. Page, “The anatomy of a large-scale hypertextual web search engine,” *Computer Networks and ISDN Systems*, vol. 30, 1998.
- [4] L. C. Freeman, “A set of measures of centrality based on betweenness,” *Sociometry*, vol. 40, no. 1, 1977.